VibeAI: evaluating image vibe representations

Research report · 30 October 2026 · decompose-then-verify with LLM judges

Results

Three ways of writing a vibe representation were compared, all generated by gpt-5: a one-line **baseline** prompt, the guided **v1** prompt, and the **workflow**, which starts from v1 and revises it with judge feedback. Each representation is split into atomic vibe claims and scored for **plausibility** (are the claims true of the image?) and **richness** (how much of what is there did it say?).

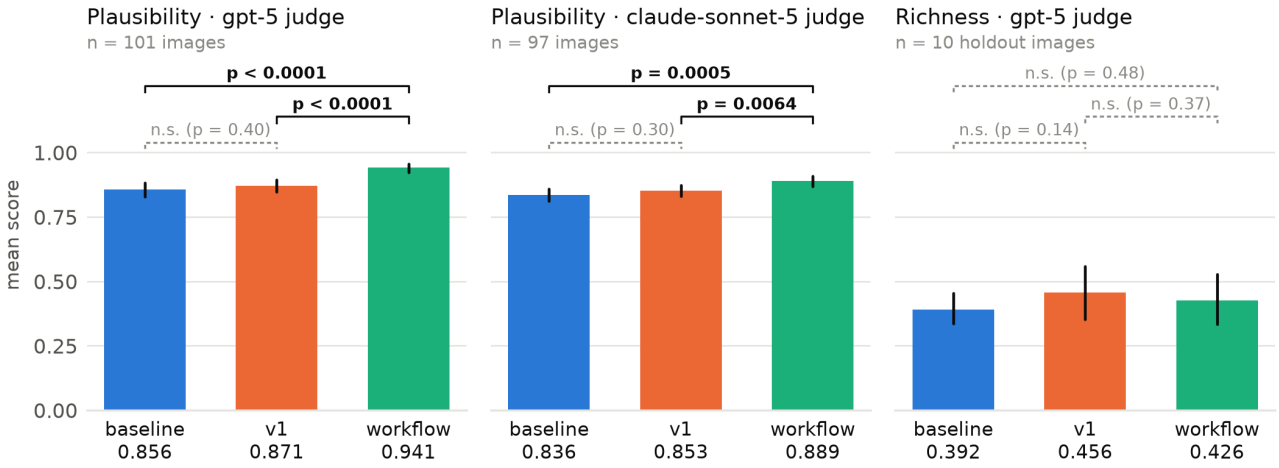


Figure 1. Mean score per system with its 95 % bootstrap interval (black line). Brackets give the paired permutation test for each pair: solid and bold = significant after Bonferroni correction ($p < 0.0167$), dashed = not significant. Richness is scored against the 4-branch vibe pool.

- **The workflow makes representations more plausible, under both judges.** With the gpt-5 judge it beats baseline by +0.085 and v1 by +0.070 (both $p < 0.0001$). With the independent Claude judge the gains shrink to +0.053 ($p = 0.0005$) and +0.036 ($p = 0.0064$). All four pass the Bonferroni threshold.
- **v1 is not more plausible than baseline** ($p = 0.40$ and 0.30).
- **Richness shows no significant differences** on 10 holdout images (all $p \geq 0.14$). v1 covers the most of the pool; the workflow covers slightly less than v1, losing most on two images where its revisions dropped vibes such as playful and adrenaline that the independent pool branches had verified.
- The gpt-5 judge shares its model with the workflow's internal gates, so part of its larger gain is likely self-preference; the Claude-judged numbers are the conservative estimate.

comparison	plausibility · gpt-5 judge	plausibility · claude judge	richness · gpt-5 judge
v1 − baseline	+0.015 (p 0.3992)	+0.017 (p 0.2981)	+0.065 (p 0.1390)
workflow − baseline	+0.085 (p < 0.0001) ✓	+0.053 (p 0.0005) ✓	+0.034 (p 0.4842)
workflow − v1	+0.070 (p < 0.0001) ✓	+0.036 (p 0.0064) ✓	-0.030 (p 0.3729)

Table 1. Mean per-image difference and two-sided p from a paired sign-flip Monte Carlo permutation test (20,000 resamples, seed 0). ✓ = below the Bonferroni threshold 0.0167 for three comparisons.

Pipeline

Every batch evaluation runs the same three steps: generate a representation from the image, decompose it into atoms, and let LLM judges score the atoms against a reference. The workflow closes the loop by feeding two of those judgements back to the generator.

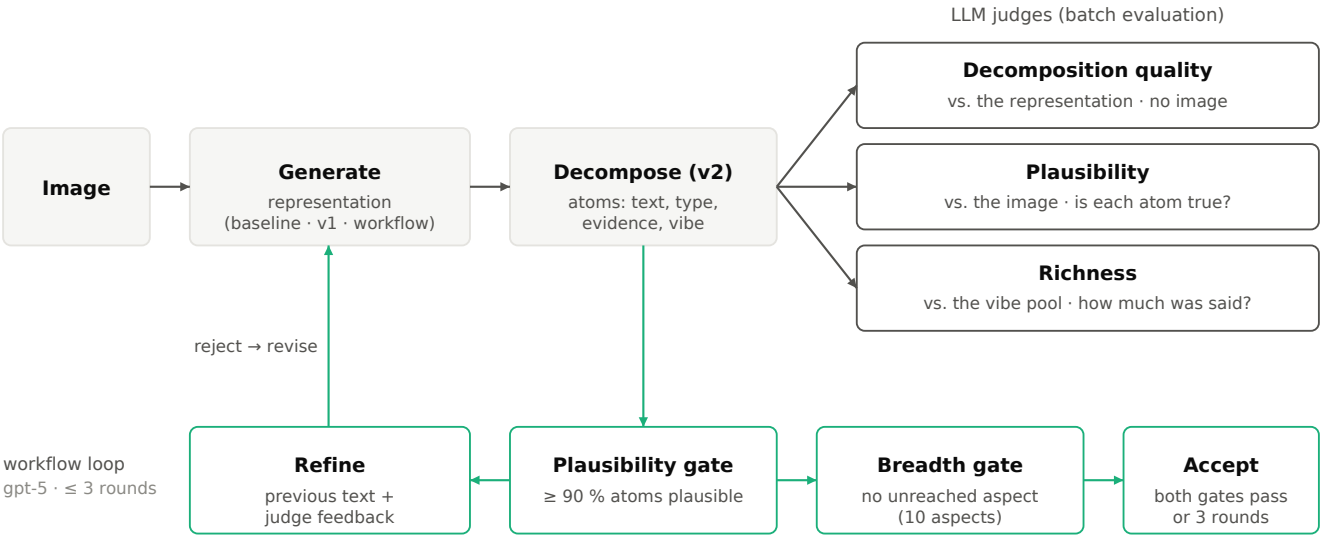


Figure 2. Batch evaluation (top) and the generator-evaluator workflow (bottom, green). In the reported runs gpt-5 generates, decomposes and judges.

Representation prompt versions

version	what it asks for	status
baseline	One line: “Describe the vibe of this image.”	compared
v1	Flowing prose that weaves evidence-backed statements (a named visual cue tied to a vibe) with vibe-only statements. Every cue must carry a vibe; vibes must be grounded and not contradicted by the image; evidence must be a main driver, not incidental; no speculation.	compared
workflow	v1 as the first draft, then a refine prompt: previous text + which claims failed and which vibes are missing. Keep what passed, re-anchor or drop failed claims, add missing aspects in a few words, keep the length. The rubric is never shown to the generator.	compared
v2	Five-step self-check in JSON: describe, split into vibe–evidence pairs, scan for contradictions, removal test for incidental evidence, final pairs.	explored
dr	Draft-and-refine in one call: draft, self-question plausibility, self-check breadth over ten vibe aspects, rewrite at the same length.	explored
rich	Describe, imagine context, then 1–3 short vibe phrases with explanations.	explored

Decomposition

A representation is split into **atoms**: minimal claims that each express one vibe. The v2 decomposer (used for all reported runs) does this in two steps and returns structured atoms.

- **Step 1, atoms.** Each atom must be affective (a mood, aesthetic or experiential feeling, not a visual fact), atomic (one interpretation; synonyms joined by a comma or “and” are split, multi-word vibe phrases kept intact), faithful (original wording, no paraphrase or added reasoning) and evidence-preserving (the exact cues, repeated when one cue supports several vibes). Together the atoms must cover the whole text.
- **Step 2, split.** Each atom is typed vibe_only or evidence_backed, and its evidence cues and its vibe are extracted verbatim. The vibe field is what richness compares.

Earlier versions: **baseline** asked for a JSON list of atomic claims; **v1** added the requirements and worked examples but returned plain sentences. The decomposition-quality judge scored baseline 0.657 → v1 0.979 → v2 0.982 (101 images; v1/v2 completeness 0.99).

Worked example (v1 representation, city_tokyo.jpg)

“…Warm lanterns strung down the street glow against the dimming sky, giving the scene a cozy, lived-in calm. The tangle of overhead wires and close, weathered facades create an intimate, slightly nostalgic urban cocoon, more whisper than bustle…”

atom (evidence_backed)	presence	direct	mapping	verdict
Warm lanterns ... give the scene a cozy calm.	✓	✓	✓	plausible
The tangle of overhead wires and close, weathered facades create an intimate urban cocoon.	✓	✓	✓	plausible
The tangle of overhead wires and close, weathered facades create more whisper than bustle.	✓	✓	✗	implausible: quiet comes from the empty street and hour, not the wires

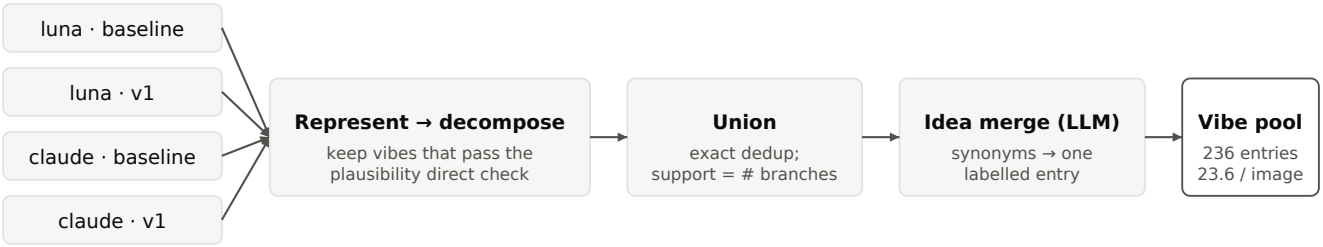
Plausibility 7/8 = 0.875. Richness for the same text was 0.29: of 31 pool vibes it covered intimate, nostalgic, quiet, lived-in and a few others, and missed cinematic, gentle mystery, melancholy and solitude.

What each judge evaluates

judge	sees	procedure	score
Decomposition quality	representation + atoms (no image)	Completeness rated 1–5. Each atom rated Good only if affective, atomic, faithful and evidence-preserving.	mean of completeness and 5 × share of Good atoms, ÷ 5
Plausibility	image + atoms	vibe_only: direct check, can the vibe be read off the image? Any contradicting cue fails it. evidence_backed, in order, stopping at the first failure: presence (is the cue there?), direct check, mapping (is the cue a main driver of the vibe?).	plausible atoms ÷ atoms
Richness	atom vibes + vibe pool (no image)	Witness test per pool vibe: can an ordinary image fit every target vibe but not this one? If yes it is distinct (missed); if no it is covered.	covered pool vibes ÷ pool size (support-weighted variant also stored)
Breadth (workflow gate)	image + atoms	For ten aspects (subject, interpersonal, object, environment, composition, colour and light, texture, style, activity, time): not applicable / covered / gap, with the missing vibe named.	covered ÷ applicable; the gate rejects on any gap

The richness reference: a vibe pool

Richness needs an answer key. For each of the 10 holdout images, four independent branches write and decompose a representation; a vibe enters the pool only if the plausibility judge's direct check accepts it. None of the branches uses gpt-5, so no compared system is scored against its own output. An LLM merge then collapses wordings of one idea (quiet / hushed / tranquil) into a single labelled entry, so an idea counts once rather than once per synonym.



4 independent branches; none is a system under test

Figure 3. Pool construction. The idea-level merge reduced the pool from 346 to 236 entries; entries found by all four branches rose from 13 to 18.

How the judges were validated

A human annotator labelled the same items as the judge in a purpose-built web app; agreement is Cohen's κ, which discounts the agreement expected by chance.

judge · labels	items	κ	raw agreement	reading
Plausibility · atom plausible?	268 atoms / 30 images	0.817	95.9 %	almost perfect
↳ evidence_backed atoms	102	0.888	96.1 %	
↳ vibe_only atoms	166	0.673	95.8 %	substantial
Decomposition · completeness (1–5)	30 images	0.696 (linear-weighted)	83.3 % exact	substantial
Decomposition · atom Good/Bad	339 atoms	0.644	82.3 %	substantial
↳ fidelity	339	0.626	82.6 %	
↳ evidence preservation	339	0.594	79.6 %	
↳ atomicity	339	0.305	96.2 %	base-rate effect*
↳ affectiveness	339	0.000	99.7 %	base-rate effect*

* Almost every atom was atomic and affective, so chance agreement is near 100 % and κ collapses even though the judge and human disagree on 1–13 atoms. Plausibility was validated on the gpt-5 judge with the baseline representation and v1 decomposition (prompt logic unchanged since). Richness, breadth and the Claude judge have not been human-validated.

Method and limitations

- Significance: paired sign-flip Monte Carlo permutation test on per-image differences, two-sided, 20,000 resamples, Bonferroni across the three comparisons. Stored in results/significance/.
- Richness rests on 10 images, so only large gaps are detectable; the pool still contains some scene descriptions (e.g. “residential Japanese district”) that are not vibes.
- Luna-bias check: two of the four pool branches use gpt-5.6-luna, a gpt-family model, which could favour the gpt-5 systems. Rescoring against a Claude-only pool (199 entries) gave no sign of that: the systems did not drop together (baseline 0.392 → 0.422, workflow 0.426 → 0.450, v1 0.456 → 0.385), and no pair differed significantly in either pool (smallest p = 0.063).
- The same check shows richness rankings at n = 10 depend on the pool: the order of the three systems changed, and the workflow's losses on two images noted in Results do not recur against the Claude-only pool.
- Agreement figures come from one annotator, and the holdout images are also part of the 101-image main set.